

Agent Based Models BootCamp

Proposal
Presentation



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**September
2024**

**“Analyzing agent's visual perception quality through
the visual perception clarity index”**

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Problem Statement

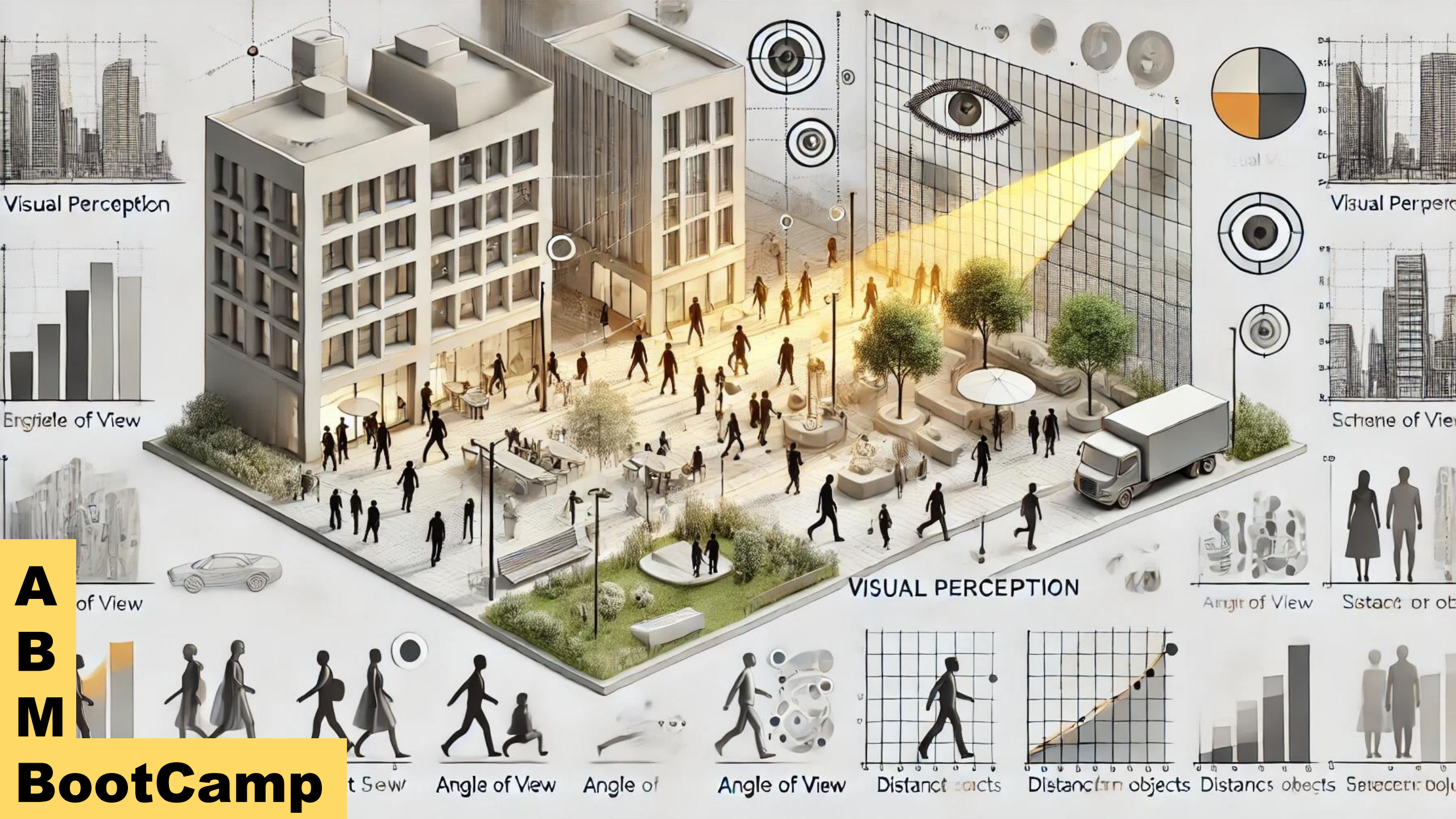
In architectural and urban spaces, user movement is influenced by various environmental factors, with visual perception being a key challenge. Visual perception is shaped by design quality, the arrangement of elements, and environmental features, and can vary based on factors such as height, angle of view, distance from objects, and physical position. As individuals navigate through spaces, their perception continuously changes.

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Visual Perception

Engriele of View

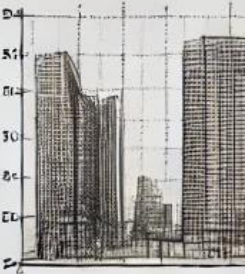
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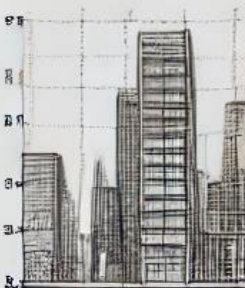
VISUAL PERCEPTION



Visual Perception



Visual Perception



Scheme of View



Static or ob

Angle of View



at Sew



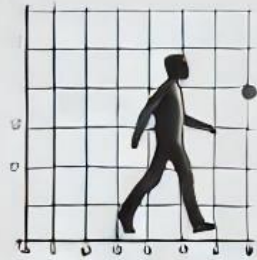
Angle of View



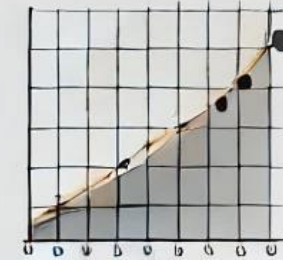
Angle of



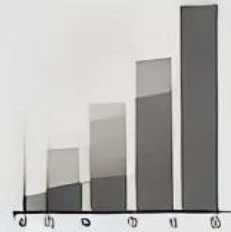
Angle of View



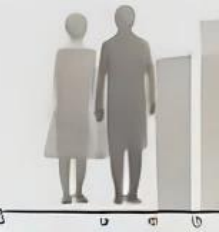
Distance objects



Distance objects



Distances objects



Distances objects

Problem Statement

Understanding these variations in visual perception is crucial for optimizing architectural and urban design. Therefore, analyzing users' visual experiences along different paths can lead to more precise and effective design outcomes.

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Research Objectives

This research aims to analyze agent' visual perception in architectural and urban spaces. The primary objective is to determine which parts of the environment and objects users observe at different points along their path, and how this observation is influenced by their distance and position.

Based on the analysis, recommendations can be made to optimize the design of public spaces, pathways, and buildings, enhancing users' visual perception and creating more effective environments.

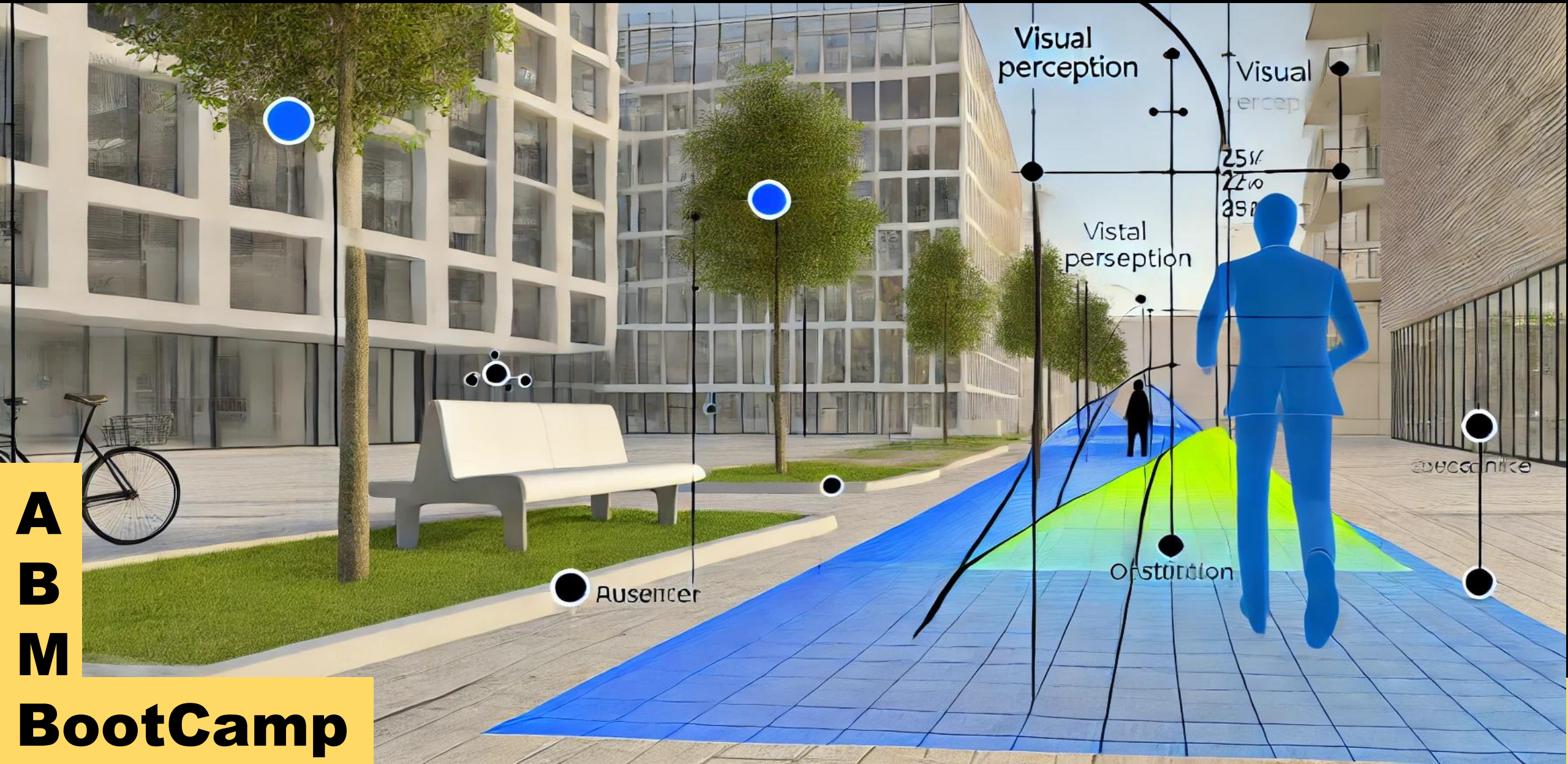
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Research Objectives



1. Geometric Environment Modeling:

The targeted urban or architectural spaces are first modeled using 3D software such as Rhino. At this stage, all objects in the environment (e.g., buildings, trees, design elements) are accurately incorporated into the model with real-world dimensions.

2. User Movement Simulation:

The agent's movement path is determined using motion simulation algorithms and path analysis within the modeled environment. At this stage, the agent's positions along the path, as well as changes in the angle of view and distance from objects, are recorded.

3. Calculation of the Visual Perception Clarity Index (VPC):

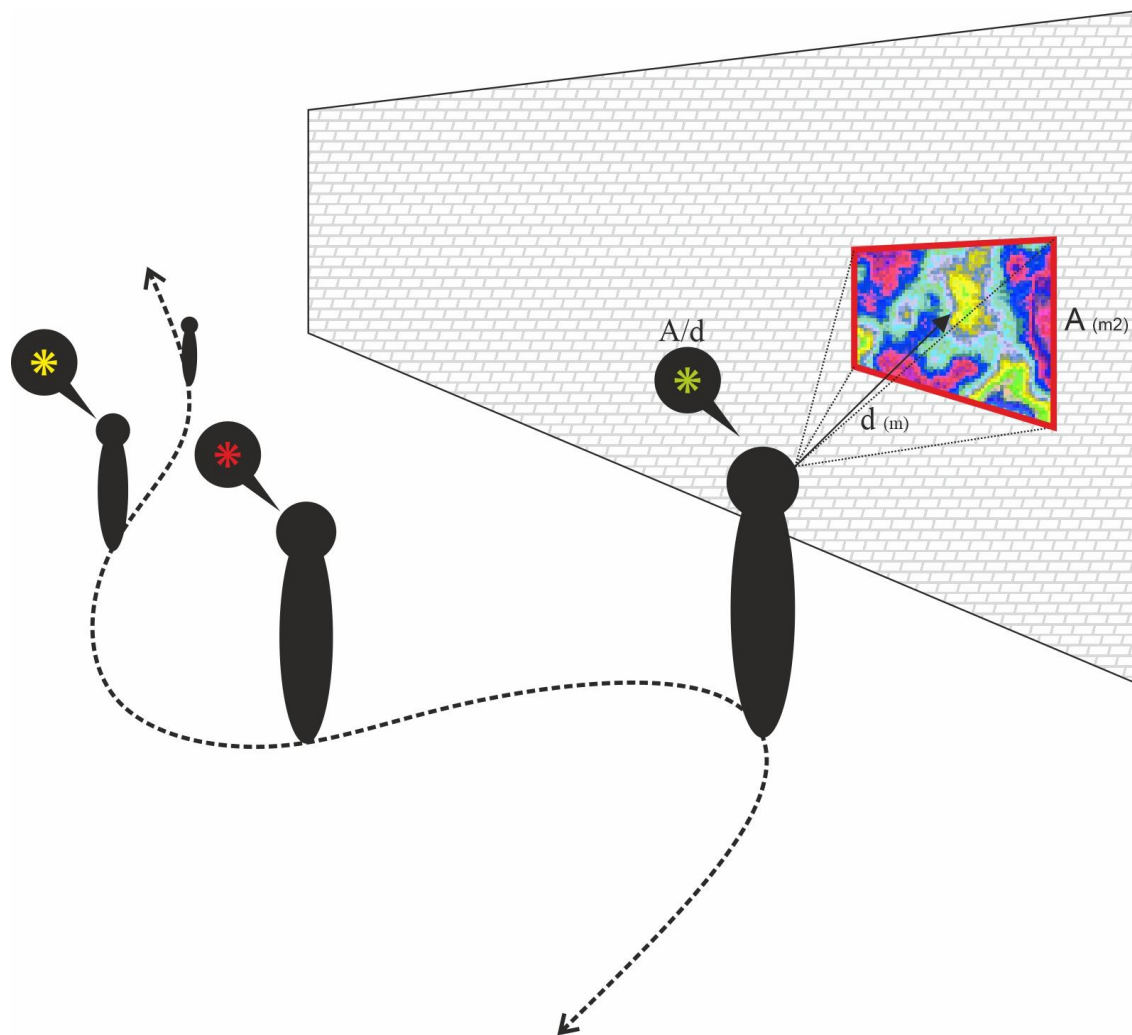
To analyze agent's visual perception, an index called VPC (Visual Perception Clarity) is defined. This index is calculated as follows:

$$A/d = VPC$$

A = the visible surface area of the object as seen by the agent (the intersection of the agent's line of sight with the object).

d = the distance from the agent to the object (the minimum length of the agent's sight lines to the surface).

Methodology



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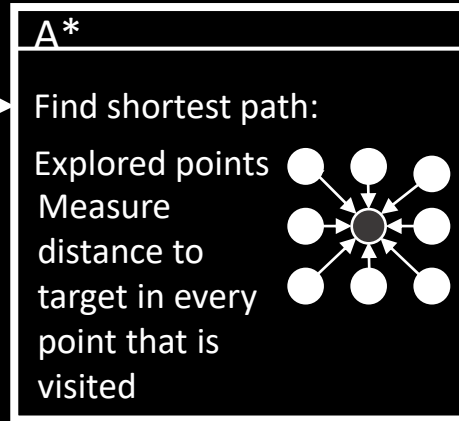
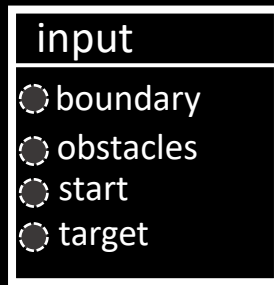


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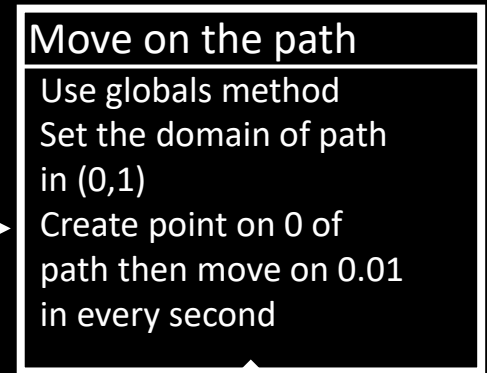
4. Analysis of Index Variations Along the Path:

Subsequently, the VPC index is calculated at various points along the path, and the data obtained is analyzed to identify which sections of the path offer the highest level of visibility. This analysis assists architects and designers in optimizing the design of spaces based on this information.

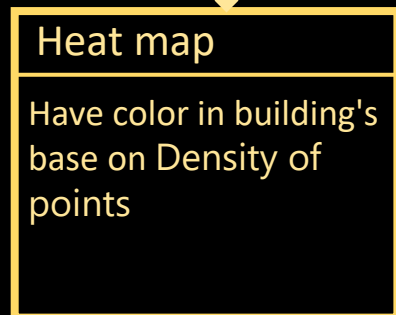
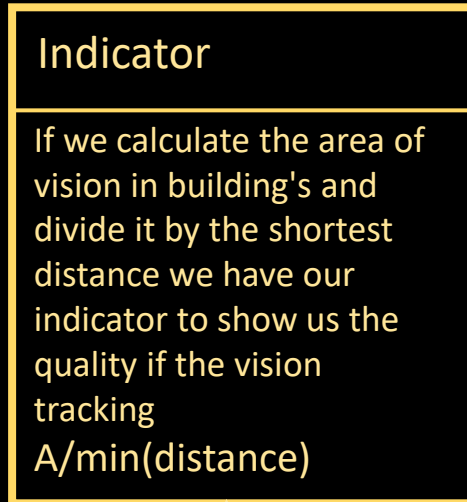
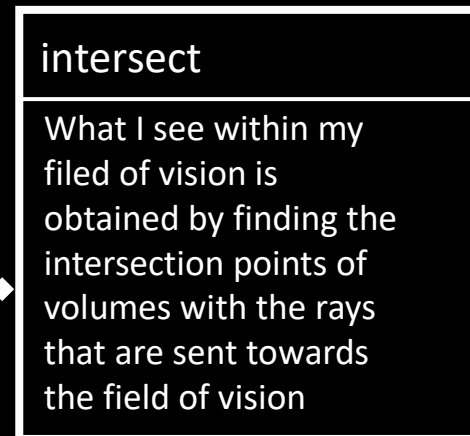
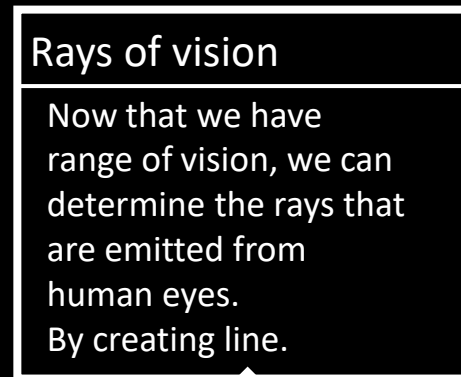
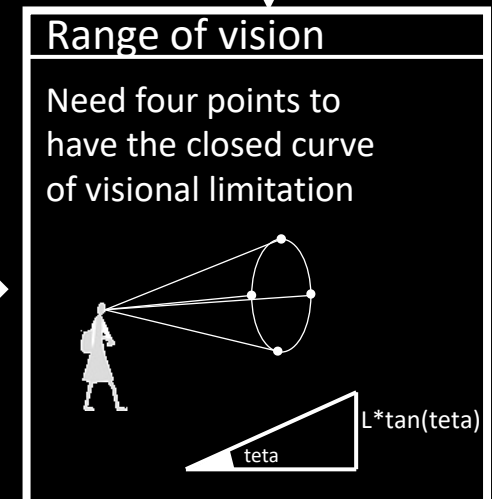
Start



Shortest path



Point in path



pseudocode

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Results

The use of the VPC index allows us to perform a precise analysis of users' visual perception in architectural and urban environments, enabling the development of optimized designs for these spaces.

Overall, the results from the proposed process can be applied in several areas:

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Results

"Because Just a Heatmap Wasn't Enough!"

To take visual perception to the next level, we've introduced a numerical index for a clearer understanding."

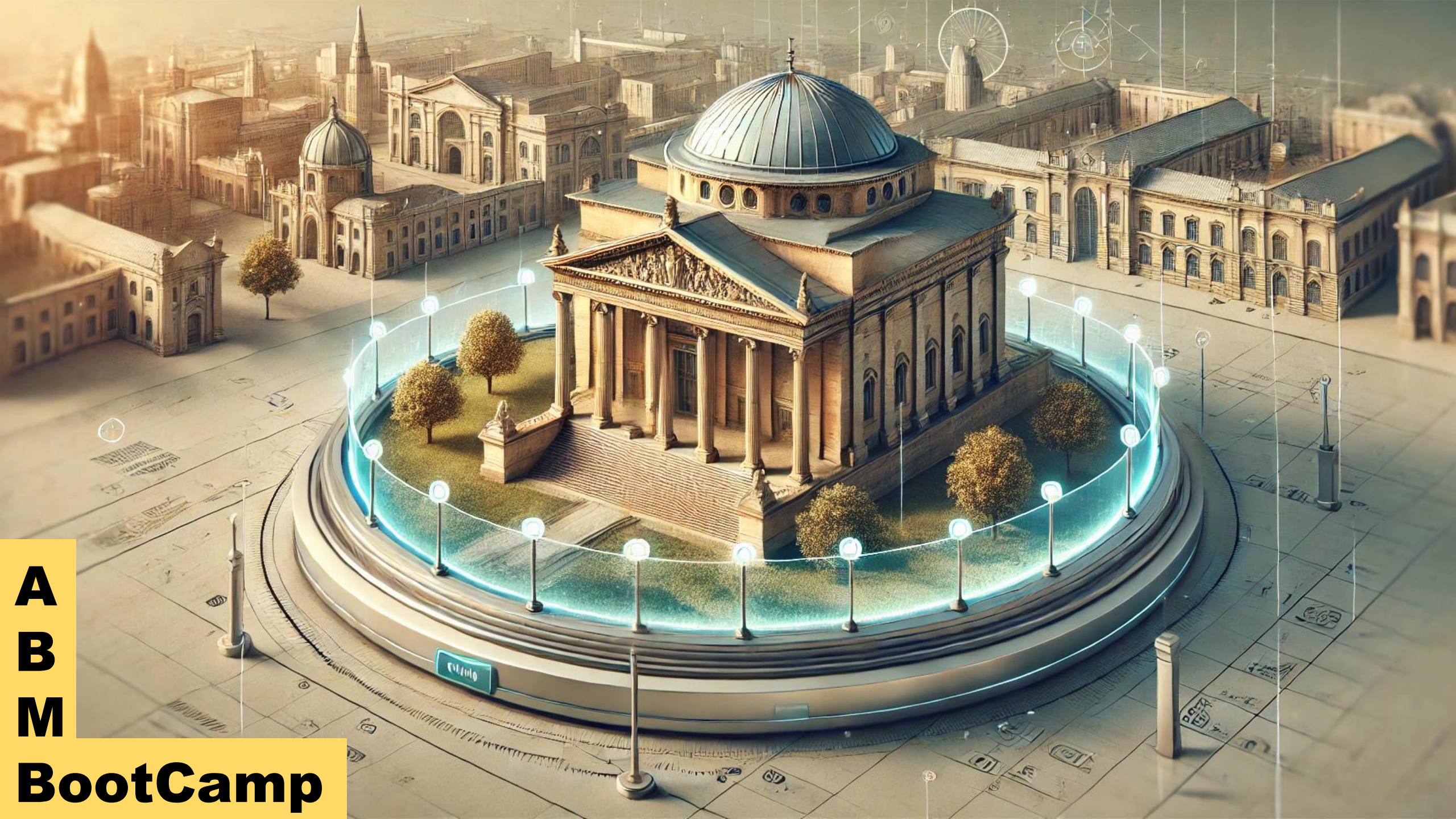
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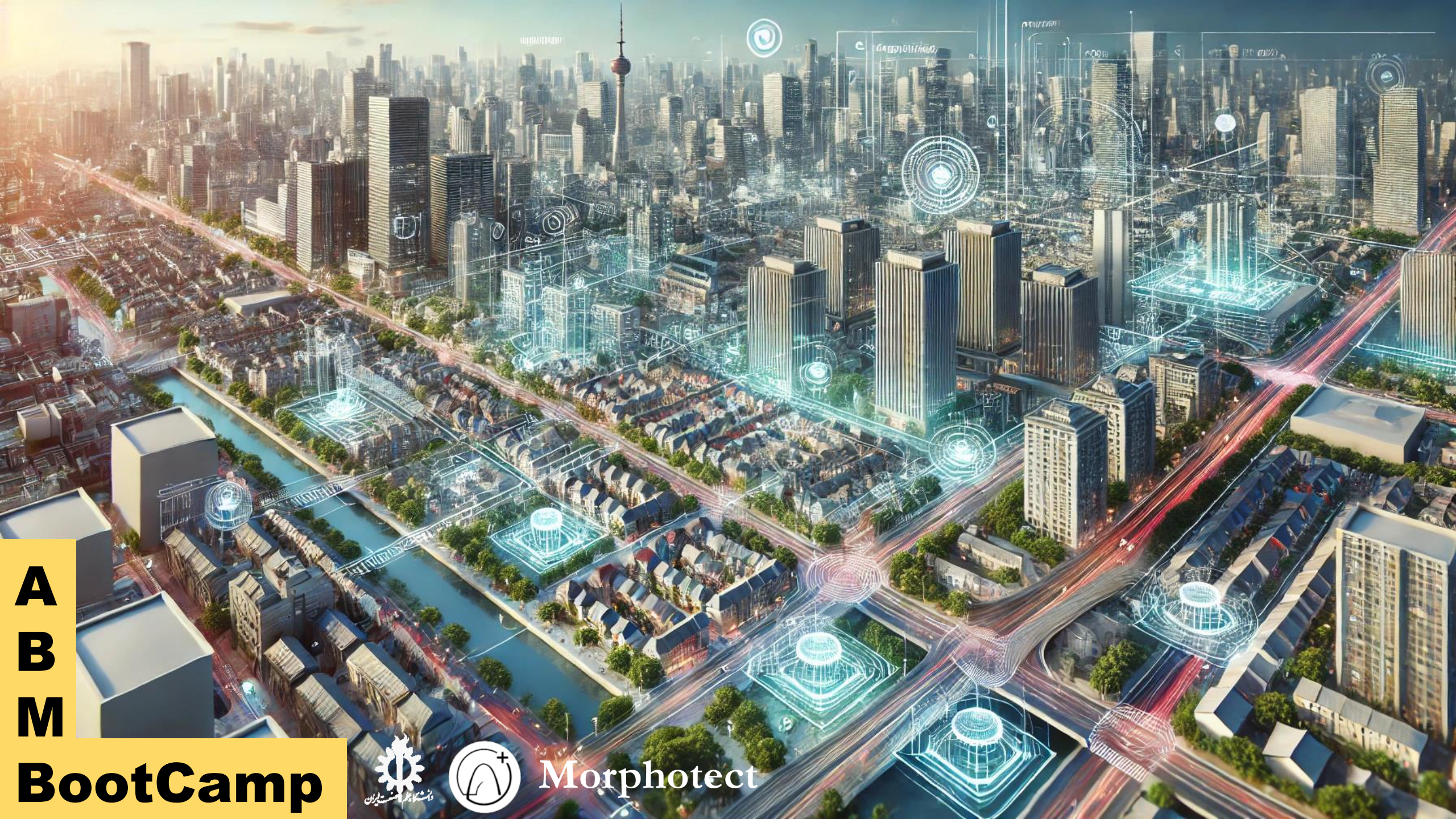
1. Defining boundaries for historical sites and buildings



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2. Spatial placement in densely populated urban areas



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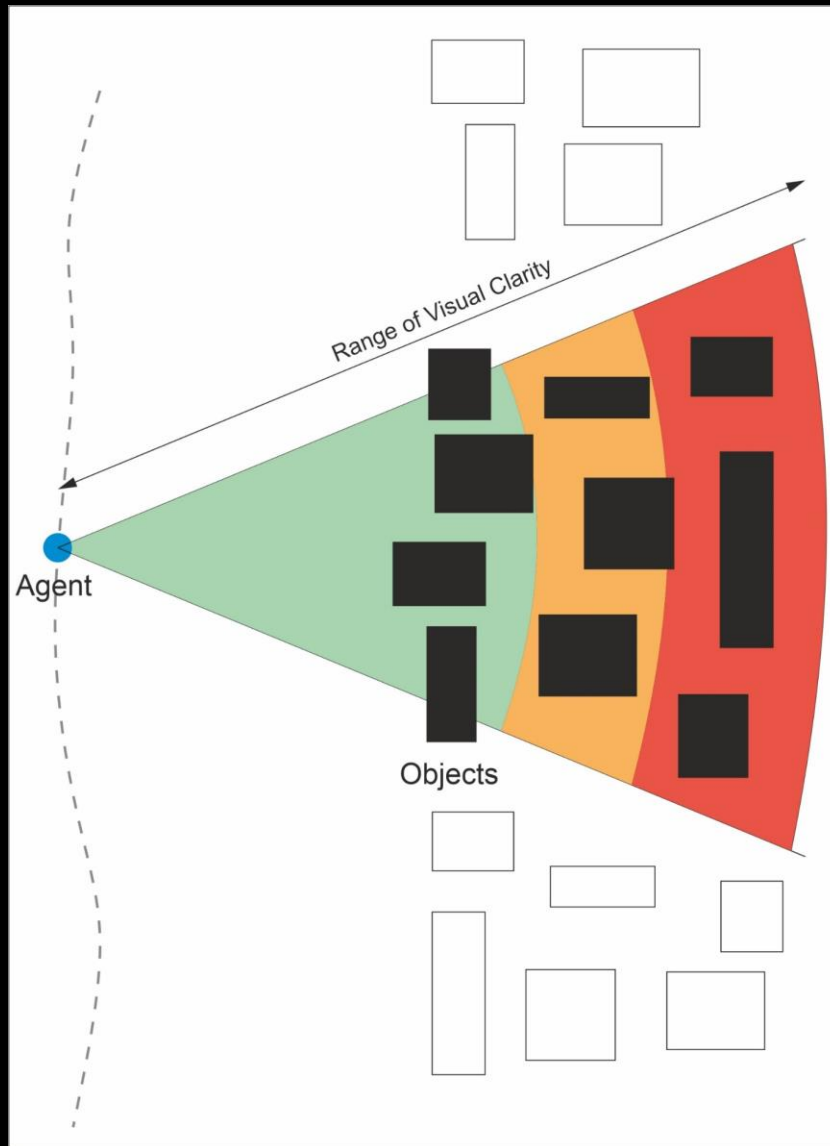


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Case Study (Parsia Residential Complex)

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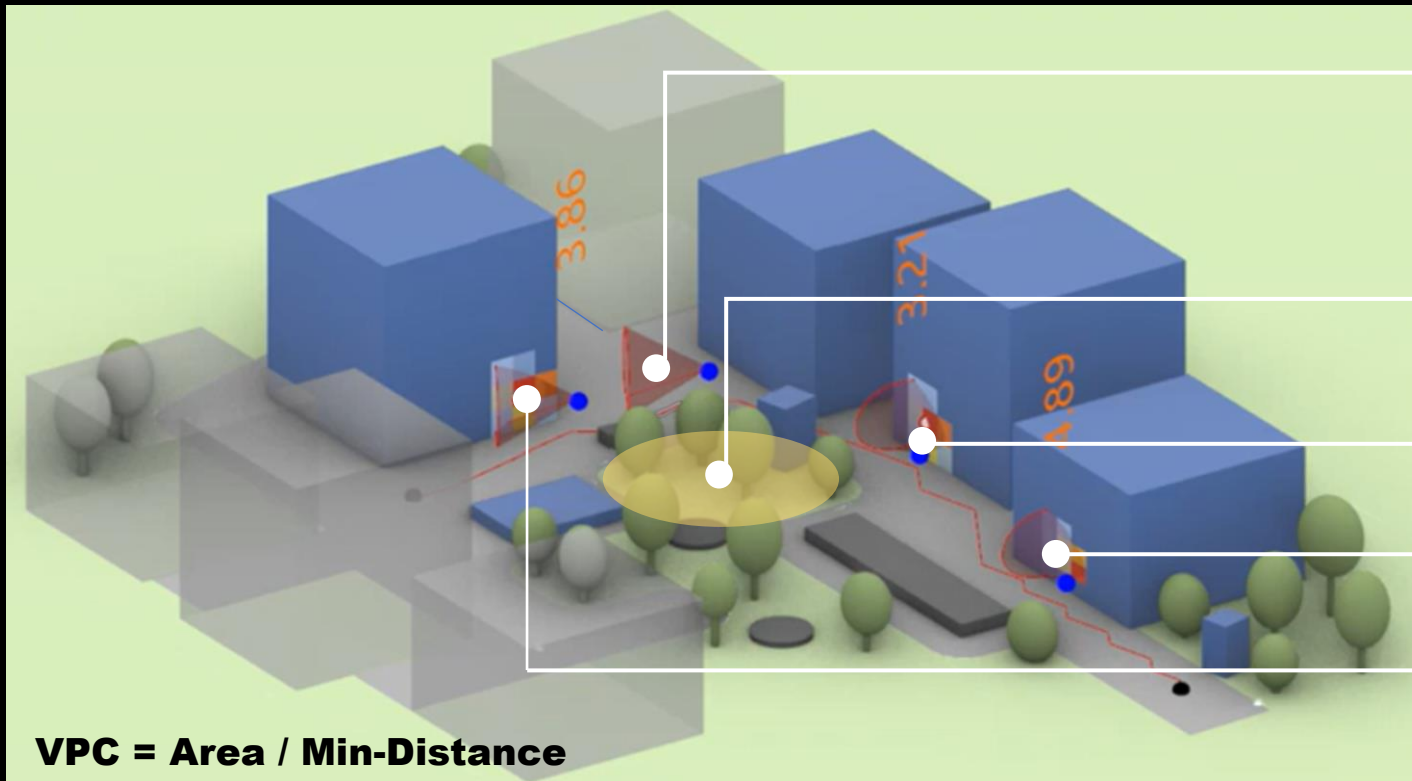


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Arranging blocks based on view quality

Identifying high-density and low density points in terms of view for site design

Designing facades using desirable and undesirable view



Adequate visibility

Potential for site design

Location with minimal view

Low security

High visibility

Limitation - Strength

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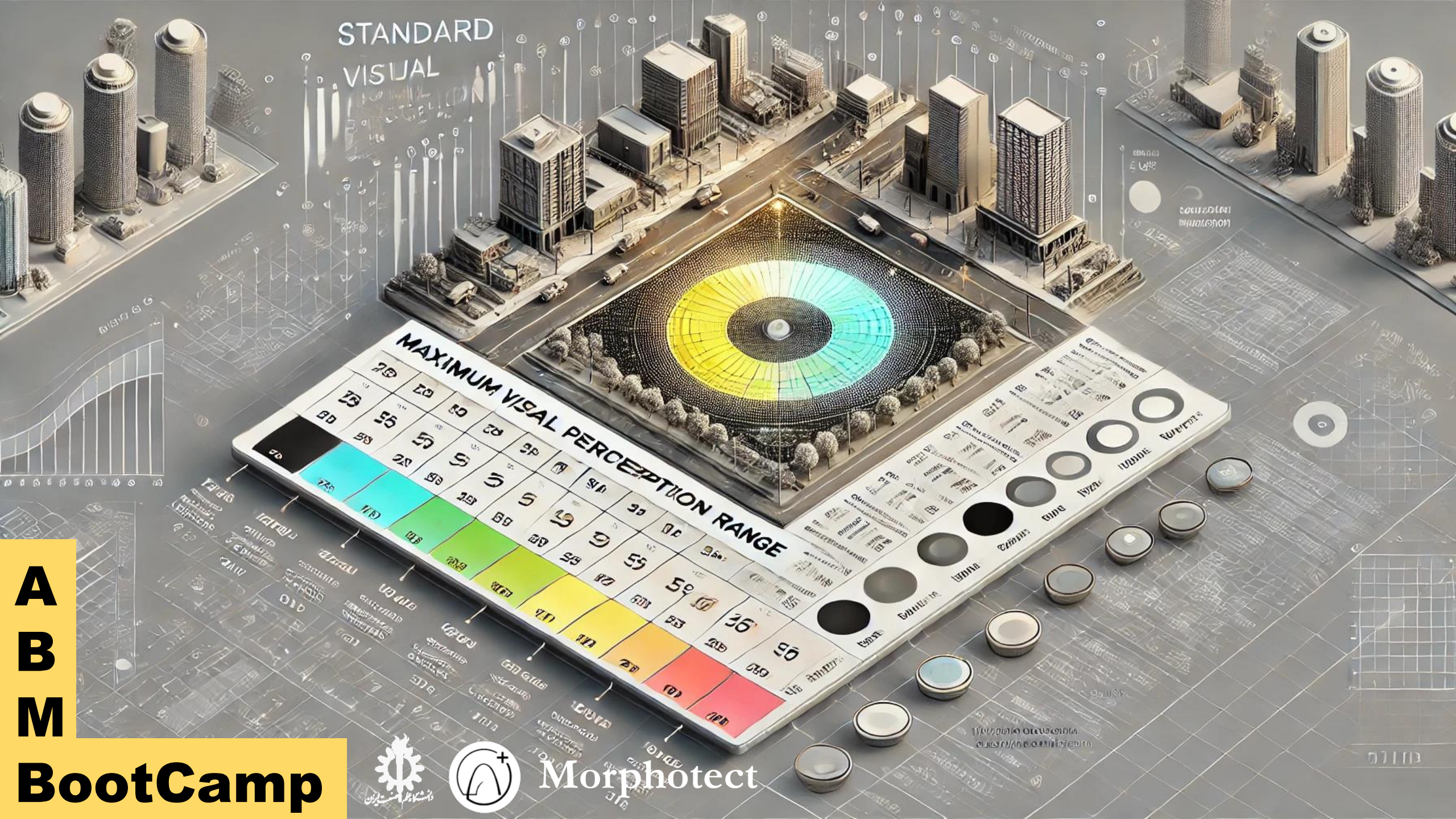
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3. Standard Table for Maximum Visual Perception Range (MVPR)

The maximum range of visibility for the agent along the path	The maximum range of visibility (on the surface)	Function	Space	Num
				1



STANDARD
VISUAL

MAXIMUM VISUAL PERCEPTION RANGE

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Limitations and Suggestions for Future Research

For Now, It's All About Shapes—Next Up:

Expanding into color ranges and even text analysis !

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Limitations and Suggestions for Future Research

“Room for Growth”

Our formula and index are just getting started—let's expand the horizons by adding more variables and refining the model !

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